

Joshua Fan

Email: jyf6@cornell.edu

Website: <http://joshuafan.github.io>

Phone: 425-628-7677

SUMMARY

PhD candidate (Cornell University) with 7+ yrs experience leading applied deep learning projects – including data engineering, understanding customer needs, research, model design, training, debugging, evaluation. Open to ML Scientist or Engineer roles where I can use my AI/ML expertise for real-world impact. Available to start in **Jan. 2026**. Prefer remote but open to relocation.

EDUCATION

Ph.D. Computer Science, Cornell University, Ithaca, NY, USA (in progress)	08/2019 – present
➤ Advisor: Prof. Carla Gomes	➤ GPA: 4.10/4.3 (Expected graduation: 12/2025)
M.S. Computer Science, University of Washington, Seattle, WA, USA	03/2017 – 06/2019
➤ Advisor: Prof. Sreeram Kannan	➤ GPA: 3.84/4.0
B.S. Computer Science, University of Washington, Seattle, WA, USA	09/2013 – 03/2017
➤ GPA: 3.97/4.0 (<i>summa cum laude</i>)	

RESEARCH EXPERIENCE

Research Assistant, Cornell University

Institute for Computational Sustainability (advised by Prof. Carla Gomes)	08/2019 – present
➤ Integrated scientific models into deep learning in an interpretable way to simulate the soil carbon cycle. Our approaches (SciReN/BINN) respect scientific laws, infer <i>unlabeled</i> biogeochemical processes, and help scientists gain insight into how input variables influence biogeochemical processes [7][8].	
➤ Designed novel self-supervised (contrastive) learning methods for satellite images to map fish farms in the Amazon. Used masked pooling and custom augmentations to overcome difficult distribution shifts , increasing F1 by 20% [5].	
➤ Created an interpretable spatiotemporal deep learning framework (GNN-RNN) to forecast crop yields in US counties [2]. With only public weather/soil data, we achieve accuracy comparable to USDA's expensive on-the-ground forecasts.	
➤ Developed novel Transformer variants which emphasize locality and relative positioning, reducing prediction error (RMSE) by an average of 18% on multiple time series regression tasks over the previous state-of-the-art [6].	
➤ Devised a weakly-supervised U-Net method, which can predict at extremely fine resolutions (30m) even when training labels are only available at a much coarser resolution (3km). We use this to predict SIF (vegetation productivity) from remote sensing data at unprecedented fine resolutions, reducing RMSE by 9% over the strongest baselines [1].	

Research Assistant, University of Washington

Information Theory Lab (advised by Prof. Sreeram Kannan)	03/2017 – 06/2018
➤ Developed scalable algorithms inspired by Latent Dirichlet Allocation and matrix factorization to discover cell types and find structure in large single-cell RNA-seq datasets (over 1 million cells) (Poster , Paper , Code)	
Computing for Development Lab (advised by Prof. Richard Anderson)	03/2015 – 06/2016
➤ Redesigned a survey app which helps health workers collect patient data and suggest follow-up actions	

INDUSTRY EXPERIENCE

Research & Development Intern, Kitware (Computer Vision team)	06/2024 – 08/2024
➤ Researched techniques to mitigate loss of plasticity in deep continual learning – to allow pretrained models to learn incrementally from new datasets and tasks	
NLP Research Intern, Docugami (AI Document Engineering startup)	06/2018 – 09/2018; 06/2019 – 08/2019
➤ Researched and implemented state-of-the-art NLP algorithms (including topic modeling, clustering, and question-answering techniques), and adapted them in novel ways for enterprise document analysis	
Software Engineer Intern, Meta (Integrity Computer Vision Team)	09/2018 – 12/2018
➤ Trained a clip-based 3D CNN to detect graphic and violent content in videos, outperforming previous approaches	
Software Engineer Intern, Meta (Search, Whole Page Ranking Team)	09/2017 – 12/2017
➤ Trained sequence neural networks to predict search behavior based on recent query history; improved user click rate	
Software Engineer Intern, Meta (Search Indexing Team)	06/2016 – 09/2016
➤ Built a web tool to help engineers debug and simulate changes to the search indexing pipeline	
Software Design Engineer Intern, BitTitan	06/2015 – 09/2015
➤ Built infrastructure to test mailbox migrations in memory; optimized SQL queries (reducing runtime by 60%)	

PUBLICATIONS (peer-reviewed conference and journal papers; * denotes equal contribution)

[1] **Monitoring Vegetation from Space at Extremely Fine Resolutions via Coarsely-Supervised Smooth U-Net.**

Joshua Fan, Di Chen, Jiaming Wen, Ying Sun, Carla Gomes.

IJCAI-22: International Joint Conference on Artificial Intelligence (AI for Good track), 2022. **(Acceptance rate: 19.3%)**

- Also presented at the “Tackling Climate Change with Machine Learning” workshop at NeurIPS 2021

[2] **A GNN-RNN Approach for Harnessing Geospatial and Temporal Information: Application to Crop Yield Prediction.**

Joshua Fan*, Junwen Bai*, Zhiyun Li*, Ariel Ortiz-Bobea, Carla Gomes.

AAAI-22: AAAI Conference on Artificial Intelligence (AI for Social Impact track), 2022. **(Acceptance rate: 15.0%)**

- Workshop version got **best paper award (ML Innovation)** at “Tackling Climate Change with ML” workshop, NeurIPS 2021.

[3] **Towards Sustainable Aquaculture in the Amazon.**

Felipe S. Pacheco, Sebastian A. Heilpern, Claire DiLeo, Rafael M. Almeida, Suresh A. Sethi, Marcela Miranda, ... , **Joshua Fan**, ... , Carla P. Gomes, & Alexander S. Flecker. *Nature Sustainability*, 2025.

[4] **Scalable preprocessing for sparse scRNA-seq data exploiting prior knowledge.**

Sumit Mukherjee, Yue Zhang, **Joshua Fan**, Georg Seelig, Sreeram Kannan. *Bioinformatics*, 2018.

WORKSHOP PAPERS (peer-reviewed)

[5] **Detecting Aquaculture with Deep Learning in a Low-Data Setting.**

L. Greenstreet, **Joshua Fan**, F. Pacheco, Y. Bai, M. Ummus, C. Doria, N. Barros, B. Forsberg, X. Xu, A. Flecker, C. Gomes.

Fragile Earth workshop at KDD 2023. [Poster](#) presented at **AGU** (American Geophysical Union) Annual Meeting, Dec 2023.

PUBLICATIONS UNDER REVIEW (* denotes equal contribution; underline denotes mentored students)

[6] **Locality and Distance-Aware Attention for Time Series Extrinsic Regression.** Submitted to AAAI 2026.

Joshua Fan*, Kaitlyn Chen*, Shufeng Kong, Junwen Bai, Zhiyun Li, Ariel Ortiz-Bobea, Carla Gomes.

[7] **Scientifically-Interpretable Reasoning Network (SciReN): Discovering Hidden Relationships in the Carbon Cycle and Beyond.** Submitted to AAAI 2026.

Joshua Fan*, Haodi Xu*, Feng Tao*, Md Nasim, Marc Grimson, Yiqi Luo, Carla P. Gomes.

[8] **Biogeochemistry-Informed Neural Network (BINN) for Improving Accuracy of Model Prediction and Scientific Understanding of Soil Organic Carbon.** Submitted to *Geoscientific Model Development (GMD)*, 2025.

Haodi Xu*, **Joshua Fan***, Feng Tao*, Lifeng Jiang, Fengqi You, Benjamin Z. Houlton, Ying Sun, Carla P. Gomes, Yiqi Luo.

[9] **AISciVision: A Framework for Specializing Large Multimodal Models in Scientific Image Classification.** *Arxiv*, 2024.

B. Hogan, A. Kabra, F. Pacheco, L. Greenstreet, **Joshua Fan**, A. Ferber, M. Ummus, A. Brito, ... , A. Flecker, C. Gomes.

SELECTED TALKS

Using Deep Learning to Monitor and Forecast Vegetation Growth.

Talk at Soil and Crop Sciences Seminar, Cornell University, Ithaca, NY, USA. Feb 2022.

TEACHING EXPERIENCE

- **TA, Cornell:** Application of Machine Learning to Plant Science (Fall 2022), Intro to Artificial Intelligence (Fall 2019)
- **TA, University of Washington:** Intro to Machine Learning (Spring 2019), Probability & Statistics (Fall 2015, Winter 2016, Spring 2017, Winter 2018, Winter 2019), Discrete Math (Fall 2016, Spring 2018)

MENTORSHIP

Kaitlyn Chen (BS Cornell 2023: next position: Amazon)

- Topic: Improved Transformer architecture with inductive biases for continuous time-series regression.

Saraswathy Amjith (high school, 1st place in Earth & Env. Sciences at WA State Science Fair – next position: MIT)

- Topic: Detecting deforestation using deep learning with radar and optical satellite imagery

TECHNICAL SKILLS

- **Significant experience:** Python, Java, C#, SQL, C++, PHP/Hack
- **Working knowledge:** R, Matlab, HTML/CSS, JavaScript, Bash
- **Tools:** PyTorch, DGL, Pandas, Matplotlib, scikit-learn, rasterio, Git, Visual Studio, xarray, Google Earth Engine, ArcGIS

HONORS & AWARDS

- **NSF Research Training (NRT) Fellowship:** Digital Plant Science (2021-2024), AI for Sustainability (2025-2026)
- **Bob Bandes Memorial Excellence in Teaching Award,** University of Washington, 2019 (one of 3 winners)